

Definition: $\forall x \in \mathbb{R}$ we let $\lfloor x \rfloor := \max \{n \in \mathbb{Z} : n \leq x\}$ so that $\lfloor x \rfloor$ is the unique integer satisfying $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$.

Quiz: State and prove the Division with remainder proposition.

Proof. $\forall a, b \in \mathbb{Z}, a > 0$

$\exists q, r \in \mathbb{Z}$ with $0 \leq r < a$ that st $b = qa + r$

From the definition: we denote $q = \lfloor \frac{b}{a} \rfloor$ and $r = b - qa$

This follows that: $b = qa + r$ and $q \leq \frac{b}{a} \leq q + 1$

We have: $qa \leq b < (q+1)a = qa + a$

Therefore: $0 \leq r < a$

basic of prove

The definition
of $\lfloor x \rfloor$

Prove of uniqueness:

for $i = 1, 2$

Assume $q_i, r_i, i = 1, 2$ that $b = q_i a + r_i$

and $0 \leq r_i < a$

proof by contradiction

We have $\lfloor \frac{b}{a} \rfloor = q_i, i = 1, 2$

since $\lfloor \frac{b}{a} \rfloor$ is the unique integer (from def)

We have $q_1 = q_2$ find confliction

$r_1 = r_2 = b - q_1 a$

Therefore, q and r are unique for $b = qa + r$